SIMATS ENGINEERING

ASSESSMENT TOOL 2: Research Review & Analysis

COURSE NAME: ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEMS COURSE CODE: MLA01

COURSE OUTCOME COVERED
CO4: *Develop intelligent software agents using suitable agent architectures and learning techniques.*(BTL 3)
Assessment Tool Weightage: 30%

Total Marks: 100 **Time: 90 Mins**
No. of Questions: 1

Annexure A

Research Review & Analysis

Code of Conduct:
I S.Venkateswarlu__ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: **S.Venkateswarlu**

Criterion	Weightage	Marks Obtained
1. Problem Analysis and Domain Understanding	10%	
2. Literature Search and Source Selection	10%	
3. Literature Review and Synthesis	15%	
4. Comparative Analysis of AI Techniques	10%	
5. Critical Analysis of Research Findings	10%	
6. Research Gap Identification	10%	
7. Proposed Research Direction	10%	
8. Research Methodology / Analysis Framework	5%	
9. Experimental / Evidence-Based Validation	10%	
10. Result Analysis and Interpretation	5%	
11. Research Paper Quality and Referencing	10%	
12. Technical Presentation and Research Defense	5%	
Total	100%	

Signature of the Faculty Total Marks Awarded

Comparative Analysis of ID3, C4.5, and CART Decision Tree Learning Algorithms: Performance, Interpretability, Complexity, and Research Directions

Abstract

Decision tree learning is one of the important approaches in supervised machine learning because it provides classification models that are relatively easy to understand and interpret. Among the classical decision tree algorithms, ID3, C4.5, and CART are widely studied and have influenced many later tree-based learning methods. This research review analyzes these three algorithms with respect to attribute-selection methods, pruning strategies, interpretability, computational complexity, and classification performance.

ID3 uses entropy and information gain to select splitting attributes, while C4.5 extends ID3 through gain ratio, handling of continuous and missing attributes, and pruning. CART uses the Gini impurity criterion for classification and follows a binary-splitting strategy with cost-complexity pruning. The reviewed literature indicates that no single algorithm is optimal for every dataset. The choice of splitting criterion, tree depth, dataset characteristics, and pruning strategy can significantly influence performance.

A major research gap is the continued dependence of classical greedy decision-tree algorithms on locally optimal splitting decisions. Recent research has shown that considering multiple candidate splits instead of only the best local split can improve accuracy while remaining more scalable than computationally expensive optimal-tree methods. Based on this observation, this paper proposes an adaptive decision-tree framework that combines multiple candidate splits, pruning, and validation-based model selection.

Keywords: Decision Trees, ID3, C4.5, CART, Information Gain, Gain Ratio, Gini Index, Pruning

1. Introduction

Artificial Intelligence and Machine Learning are increasingly used for classification and prediction problems in areas such as healthcare, education, finance, agriculture, cybersecurity, and business. A machine learning model should not only provide accurate predictions but should also be understandable when the application requires transparency.

Decision trees are particularly useful because their decisions can be represented as a sequence of tests and rules. A path from the root node to a leaf can be interpreted as an IF-THEN decision rule. This makes decision trees attractive for applications where model interpretability is important.

Three classical algorithms are particularly important in the development of decision-tree learning:

- ID3
- C4.5
- CART

ID3 was developed around the concept of selecting attributes using information gain. C4.5 extended this approach by addressing several practical limitations of ID3, including continuous attributes, missing values, and pruning. Quinlan's C4.5 work specifically describes constructing decision trees, handling unknown attribute values, pruning, and converting trees into rules. (ScienceDirect)

CART, introduced by Breiman and colleagues, developed a tree-based methodology using splitting rules, classification and regression procedures, and pruning techniques. The CART methodology includes Gini-based splitting for classification and cost-complexity considerations for selecting tree size. (Google Books)

This review compares these approaches and investigates their advantages, limitations, and possible research improvements.

2. Research Problem and Objectives

1. Research Problem

Although ID3, C4.5, and CART are well-established decision-tree algorithms, they differ in their splitting criteria, tree construction strategies, handling of data, pruning methods, computational requirements, and predictive behavior.

The main research problem is:

How do ID3, C4.5, and CART differ in attribute selection, pruning, interpretability, computational complexity, and classification performance, and how can their limitations be addressed through improved decision-tree learning?

2.2 Objectives

The objectives of this research review are:

1. To study the principles of ID3, C4.5, and CART.

2. To compare their attribute-selection methods.
3. To analyze their pruning strategies.
4. To compare their interpretability.
5. To analyze their computational complexity.
6. To examine their classification performance.
7. To identify strengths and weaknesses of each algorithm.
8. To identify limitations and research gaps in classical decision-tree learning.
9. To examine recent improvements to decision-tree learning.
10. To propose a possible future research direction.

3. Literature Review

1. ID3

ID3, or Iterative Dichotomiser 3, is one of the classical decision-tree learning algorithms. It constructs a tree using a top-down recursive strategy.

The main principle of ID3 is to select the attribute that provides the highest information gain.

Information gain is based on entropy.

The entropy of a dataset can be represented as:

$$\textbf{Entropy}(S) = -\Sigma\, p_i \log_2(p_i)$$

where p_i represents the proportion of examples belonging to class i.

Information gain can be expressed as:

$$\textbf{Gain}(S,A) = \textbf{Entropy}(S) - \Sigma_v\, (|S_v| / |S|)\, \textbf{Entropy}(S_v)$$

The attribute with the highest information gain is selected for splitting.

Advantages of ID3

- Simple to understand.
- Easy to implement.
- Produces interpretable decision trees.
- Works well for categorical attributes.
- Provides understandable decision rules.

Limitations of ID3

- Information gain can favor attributes with many distinct values.
- Classical ID3 is primarily designed for categorical attributes.
- It does not provide the more advanced pruning mechanisms found in C4.5.
- Large trees can overfit training data.

Research comparing ID3 and C4.5 describes C4.5 as an extension of ID3 that adds capabilities such as continuous attributes, missing-value handling, and pruning. (SAI Organization)

2. C4.5

C4.5 was developed by J. Ross Quinlan as an extension of the earlier ID3 approach.
Instead of using information gain alone, C4.5 uses the **Gain Ratio** to select splitting attributes.
Gain ratio can be expressed as:
Gain Ratio = Information Gain / Split Information
This reduces some of the bias that information gain can have toward attributes containing many distinct values.
C4.5 also provides support for:

- Continuous attributes
- Missing values
- Tree pruning
- Rule generation
- Classification using decision trees

Quinlan's description of C4.5 includes explicit treatment of continuous attributes, unknown attribute values, error-based pruning, and converting trees into rules. (Elsevier Shop)

Advantages of C4.5

1. Handles categorical and continuous data.
2. Can handle missing attribute values.
3. Uses gain ratio.
4. Provides pruning.
5. Produces interpretable rules.
6. Can reduce overfitting through pruning.

Limitations of C4.5

1. Tree construction can become computationally expensive for large datasets.
2. The resulting tree may still become complex.
3. Performance depends on the dataset and parameter choices.
4. Greedy splitting can lead to locally optimal rather than globally optimal trees.

3.CART

CART stands for **Classification and Regression Trees**.

The CART methodology was developed by Breiman and collaborators and supports both classification and regression tasks. The original CART work presents tree construction, splitting rules, pruning, and statistical analysis of tree-based models. (Google Books)

For classification, CART commonly uses the **Gini impurity** criterion.

Gini impurity is:

Gini(S) = $1 - \Sigma p_i^2$

A lower Gini impurity indicates a more homogeneous node.

CART generally uses binary splits, meaning an internal node is divided into two branches.

Advantages of CART

1. Can be used for classification and regression.
2. Uses a simple binary-splitting strategy.
3. Can handle numerical variables effectively.
4. Uses pruning to control tree complexity.
5. Produces interpretable models.
6. Can be used for a wide range of datasets.

Limitations of CART

1. A deep tree can overfit.
2. Small changes in data can sometimes produce different trees.
3. Binary splitting can result in deeper trees.
4. Performance depends on splitting and pruning choices.

4. Comparative Analysis

The three algorithms can be compared using the following parameters.

Parameter	**ID3**	**C4.5**	**CART**
Main criterion	Information Gain	Gain Ratio	Gini Impurity
Splitting	Multiway	Multiway	Binary
Continuous attributes	Limited in original form	Supported	Supported
Missing values	Limited	Supported	Can handle through tree mechanisms
Pruning	Limited/basic	Error-based pruning	Cost-complexity pruning
Classification	Yes	Yes	Yes

Parameter	ID3	C4.5	CART
Regression	No	Primarily classification	Yes
Interpretability	High	High	High
Tree complexity	Can become large	Reduced through pruning	Controlled using pruning
Main advantage	Simple	Flexible	Classification and regression
Main limitation	Bias toward many-valued attributes	More computational complexity	Binary trees may become deep

The comparison is consistent with research describing ID3, C4.5, and CART as classical tree-learning strategies using different splitting criteria. A 2023 comparative study evaluated these and additional decision-tree methods using classification accuracy, tree depth, leaf nodes, and construction time. (Sage Journals)

1. Attribute Selection

ID3

ID3 selects the attribute with the highest information gain.

Criterion:

Information Gain

C4.5

C4.5 uses gain ratio.

Criterion:

Gain Ratio

This attempts to reduce the bias toward attributes with many possible values.

CART

CART uses Gini impurity for classification.

Criterion:

Gini Index

Therefore, the main difference between the three algorithms is the mathematical criterion used to determine the best split.

2. Pruning Strategies

Pruning is important because an excessively large tree can memorize training data and perform poorly on unseen data.

ID3

The original ID3 approach has limited pruning compared with later algorithms and can produce unnecessarily large trees.

C4.5

C4.5 incorporates error-based pruning and can simplify the decision tree after construction. (Elsevier Shop)

CART

CART uses cost-complexity pruning to balance tree accuracy and tree size. The original CART methodology explicitly discusses right-sized trees and pruning. (Google Books)

3. Interpretability

All three algorithms have an important advantage over many complex machine-learning approaches: their decisions can be represented as rules.

For example:

IF attendance > 75 AND internal marks > 60 THEN Pass

Such rules can be understood by teachers, administrators, business users, and domain experts. C4.5 particularly supports conversion from decision trees into IF-THEN rules. (Elsevier Shop)

4. Computational Complexity

Decision-tree construction requires evaluating candidate attributes and possible splits.

In general:

- ID3 can be relatively simple for small categorical datasets.
- C4.5 requires additional calculations for gain ratio, continuous values, missing values, and pruning.
- CART evaluates binary splits and performs pruning procedures.

Therefore, computational cost depends not only on the algorithm but also on:

- Number of samples
- Number of features
- Number of possible attribute values
- Number of candidate thresholds

- Tree depth
- Pruning strategy

For practical applications, the trade-off between predictive performance and computational cost must therefore be considered.

5. Critical Analysis

The literature shows that decision trees remain useful because of their interpretability and relatively straightforward learning process.

However, each algorithm has limitations.

ID3 – Critical Analysis

Strengths

- Simple algorithm.
- Easy to explain.
- Easy to implement.
- Effective for categorical datasets.

Weaknesses

- Information-gain bias.
- Limited original support for continuous data.
- Can produce large trees.
- Can overfit.

C4.5 – Critical Analysis

Strengths

- More flexible than ID3.
- Gain ratio improves attribute selection.
- Handles continuous data.
- Handles missing values.
- Includes pruning.
- Produces interpretable rules.

Weaknesses

- More complex than ID3.
- Can require more computation.
- Still follows a greedy tree-building strategy.

- Tree performance can vary between datasets.

CART – Critical Analysis

Strengths

- Supports classification and regression.
- Uses binary splitting.
- Gini criterion is computationally convenient.
- Provides pruning.
- Applicable to numerical data.

Weaknesses

- Can produce deep trees.
- Sensitive to training data changes.
- A single tree may have limited predictive power compared with ensembles.
- Greedy splitting can miss globally better tree structures.

6. Research Gap

The major research gap identified from the review is the **greedy nature of classical decision-tree learning**.

ID3, C4.5, and CART generally make a locally optimal splitting decision at each stage. Once a split has been selected, the algorithm continues recursively.

This approach is computationally practical, but the locally best split is not always part of the globally best tree.

Recent research has investigated alternatives to selecting only one best split. Blanc et al. proposed a **Top-k** approach in which multiple high-quality candidate attributes are considered rather than only the single best attribute. Their theoretical and experimental results indicate that considering additional candidate splits can improve accuracy while remaining more scalable than computationally expensive optimal decision-tree approaches. (arXiv)

Another research direction is the development of improved splitting criteria. A 2023 study compared six decision-tree strategies using accuracy, tree depth, number of leaves, and construction time, showing that the choice of splitting criterion can significantly influence model behavior. (DOI)

Therefore, the research gaps include:

1. Dependence on greedy split selection.

2. Possible instability caused by small data changes.
3. Difficulty balancing accuracy and interpretability.
4. Potential overfitting in complex trees.
5. Need for efficient methods that consider multiple candidate splits.
6. Need for adaptive pruning based on validation performance.
7. Need for better comparison across different datasets and application domains.

7. Proposed Research Direction

Based on the identified research gap, an **Adaptive Multi-Split Decision Tree** can be proposed.

Proposed Approach

Instead of selecting only the single best attribute at each node, the proposed approach can consider the top few candidate splits.

Proposed Framework

Dataset

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Data Preprocessing

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Candidate Attribute Generation

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Calculate Split Scores

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Select Top-k Candidate Splits

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Evaluate Candidate Subtrees

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Choose Best Validated Split

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Tree Construction

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Adaptive Pruning

↓

Final Decision Tree

↓

Prediction

Main Components

1. Candidate Split Generation

Calculate information gain, gain ratio, or Gini-based scores for candidate attributes.

2. Top-k Selection

Instead of selecting only one attribute, retain the top k candidate splits.

3. Validation-Based Selection

Evaluate candidate splits using validation performance rather than relying only on local impurity reduction.

4. Adaptive Pruning

Remove branches that do not improve validation performance.

5. Interpretability Constraint

Keep the tree within a specified maximum depth or complexity level.

Expected Benefits

The proposed method is expected to:

- Improve classification accuracy.
- Reduce dependence on one greedy choice.
- Control overfitting.
- Maintain interpretability.
- Provide a balance between accuracy and computational cost.

The Top-k decision-tree research provides evidence that considering several candidate splits can improve performance while remaining substantially more scalable than fully optimal decision-tree approaches. (arXiv)

8. Discussion

The comparison shows that ID3, C4.5, and CART represent different stages in the development of decision-tree learning.

ID3 provides a simple foundation based on information gain. C4.5 improves this framework by introducing gain ratio, continuous-value processing, missing-value handling, and pruning. CART provides a different design based on binary splits, Gini impurity for classification, and cost-complexity pruning.

No algorithm can be considered universally superior.

For simple categorical datasets, ID3 can be sufficient. For datasets containing continuous attributes and missing values, C4.5 provides additional flexibility. For applications involving both classification and regression, CART is particularly useful.
The choice of algorithm should therefore depend on:

- Dataset characteristics
- Number of features
- Number of classes
- Data types
- Missing values
- Required interpretability
- Computational resources
- Accuracy requirements
- Need for classification or regression

Research on different splitting criteria further demonstrates that split selection can influence accuracy, tree depth, leaf count, and construction time. (DOI)
The major insight is that decision-tree research should not focus only on maximizing training accuracy. A good decision-tree model should achieve a balance between:
Accuracy + Generalization + Interpretability + Computational Efficiency

9. Conclusion

This research review analyzed three classical decision-tree algorithms: ID3, C4.5, and CART. ID3 uses information gain for attribute selection and provides a simple approach to tree construction. C4.5 extends ID3 by introducing gain ratio, handling continuous and missing attributes, and applying pruning. CART uses Gini impurity for classification, binary splitting, and cost-complexity pruning.
The comparison shows that all three algorithms provide interpretable models, but they differ in flexibility, computational requirements, splitting criteria, and pruning mechanisms.
The major research gap identified is the greedy nature of classical decision-tree construction. Selecting only the locally best split may prevent the algorithm from finding a better overall tree. Recent research on Top-k decision-tree learning suggests that considering multiple candidate splits can improve accuracy while maintaining scalability. (arXiv)
Therefore, future research can focus on adaptive multi-split decision trees that combine multiple candidate attributes, validation-based selection, and adaptive pruning. Such an

approach could provide a better balance between predictive performance, generalization, interpretability, and computational efficiency.

References

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